

Aakash Rajan

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EDUCATION

Indian Institute of Technology, Madras

Bachelor of Technology, Mechanical Engineering

CGPA: 9.01 / 10

Aug 2024 – Present

Head Start Educational Academy

ISC, Class XII

Percentage: 96%

2024

SCHOLASTIC ACHIEVEMENTS AND COMPETITIONS

- **JEE Advanced 2024:** Secured a rank of **2130** out of 180k+ students, placing in the **top 1.5% of students across India**.
- **Caterpillar Tech Challenge:** Led a group of 3, securing **1st place** in Caterpillar's national contest with **7500+ competitors**.
- **Formula Bharat 26:** Placed **2nd** across top **Formula Student** teams across India, as a part of Team **Raftar Formula Racing**.

RESEARCH AND TECHNICAL EXPERIENCE

Research Intern – Video World Models

Jun 2026 – Present

HiRo Lab, Indian Institute of Science (IISc) – Advisor: Prof. Ravi Prakash

Bengaluru, India

- * Working on making video **world models** more **geometry-aware** for closed-loop robot navigation; modifying, training, and running inference on models including the Gaussian World Model, **DINOv3**, and $\pi 0.5$.
- * Devised a **memory-retrieval** mechanism for WorldPack, improving **SSIM** by $\sim 3.5\%$ and **LPIPS** by $\sim 3\%$ over baseline on the LoopNav benchmark.
- * **Prototyped** RoPE-based trajectory packing and geometric/**semantic attention conditioning** (using DINOv3) on the Navigation World Model (NWM) backbone.

Undergraduate Researcher – Multi-Agent RL for Air Traffic Control

Aug 2026 – Present

Indian Institute of Technology Madras – Advisor: Prof. Rahul Meshram

Chennai, India

- * Participating in **TU Delft's RL competition** for air traffic control, evaluated on **safety, efficiency, and goal completion**.
- * Developing **RL agents** in TU Delft's **BlueSky-Gym air-traffic environment** for safe navigation, avoiding static and moving obstacles.
- * Designing a **potential based reward function**; Implemented **Q-learning**, SARSA for Single agent RL; Testing **MLPs**, **GNN** feature encoding, value function estimates and priority sorting for collision avoidance.

Multi-Agent Autonomous Dump-Truck Simulator

Mar 2026 – Jun 2026

Caterpillar National Tech Challenge - Winner

India

- * Led a team of 3 to develop a **multi-agent** dump-truck simulator for **autonomous** dump packing on python, winning Caterpillar's national contest among **7,500+ competitors**.
- * Designed and implemented a **novel** algorithmic framework to integrate discretized **World Representation**, a 6-term heuristic and **Monte Carlo Tree Search (MCTS)** for long horizon planning and **Combinatorial Optimization** using the **Hungarian Algorithm**.
- * Integrated **vehicle dynamics** and **MAPF** using **Hybrid A*-CBS** to optimize complex truck reverse maneuvers into dump targets.

Hub Motor Cooling Jacket Design

Jul 2025 – Nov 2025

Formula Student, Raftar Formula Racing

Chennai, India

- * Designed a **lattice-strut based cooling jacket** for an electric hub motor, optimizing pressure drop and heat-transfer coefficient backed by **real life testing** of an LPBF manufactured prototype.
- * Developed a MATLAB iterative solver for **cooling-loop head loss**, coupled with multi-phase CFD thermal analysis to **improve simulation accuracy**.

Biometric Medicine Dispenser

Apr 2026 – May 2026

Mechatronics Course Project (ME2400), IIT Madras – Advisor: Prof. Sushanta Panigrahi

Chennai, India

- * Led a team of 6; Engineered the embedded control architecture (**UART, I²C**), implemented **state-machine logic** for authenticated dispensing, missed-dose recovery, refill management, and sensor-actuator coordination.
- * Implemented a closed loop system to detect dispensing using **IR sensing** and **load-cell feedback**.
- * Awarded **Best Project in Batch** by the former HOD, Prof. Chandramouli.

TECHNICAL SKILLS

Languages: Python, Java, MATLAB

Frameworks & Robotics: PyTorch, ROS2, Reinforcement Learning, World Models, AI/ML Implementation

Softwares: AutoCAD, Fusion360, FEA, Ansys CFD, Simulink

RELEVANT COURSEWORK

- **Robotics and Control:** Measurements, Instrumentation and Control | Basic Electrical Engineering
 - **Thermal and Fluids:** Applied Thermodynamics | Fluid Mechanics | Heat Transfer*
 - **Mechanics and Design:** Kinematics and Dynamics of Machinery | Machine Design* | Materials and Design
 - **Mathematics and Programming:** Linear Algebra | Differential Equations | Complex Networks* | Python Data Structures & Algorithms* | Intro to Scientific Computing (MATLAB)
- * - Ongoing

POSITIONS OF RESPONSIBILITY

Aerodynamics and Cooling Engineer
Formula Student, Raftar Formula Racing

Apr 2025 – Dec 2025
IIT Madras

- * Part of the engineering team responsible for aerodynamic component design and manufacturing; gained hands-on knowledge of aerodynamics, cooling strategies, vehicle dynamics & power electronics integration.
- * Designed, tested and manufactured the **aerodynamic components** of the car through **Ansys simulations**, tufts, analytical analysis

Music Contingent Head (Piano)
Inter-IIT Cultural Meet 8.0, IIT Madras

July 2026 – Present
IIT Madras

- * Selected a **20 member** music team to represent **IIT Madras** for **InterIIT 2026** amongst 500+ applicants
- * Proposed a semesterly budget for the Music Club of upto **2.7 Lakhs** approved by the IITM administration; **Achieved improved quality** of club performances with IEMs purchase, new mixer console setup.
- * Led a team of 8 musicians as **Music Director** of Stagecoach, Drama Club’s highest-budget annual event.
- * Represented IIT Madras at Inter-IIT Cultural Meet 8.0 as **Lead Keyboardist**, placing 4th among 23 IITs.
- * Completed all 8 grades by **ABRSM** with merit in performance piano; Performed in several stages across India.

Business Plan Presentation – Runners Up
Raftar Formula Racing, Formula Bharat 2026

Jun 2025 – Sept 2025
IIT Madras

- * Developed a fully fledged business plan for **Formula Bharat 2026**, securing **2nd place** out of 40+ teams *across India*.
- * Worked on market analysis, **TCO analysis**, competitive analysis, **supply chain**, and CAC estimation; developed a pitch deck and finance sheets for the fictional company.